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Implementing Online Analytical Processing in Hotel Customer Relationship Management

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Abstract. Online Analytical Processing (OLAP) is increasingly being used by applying technology concepts that use a multidimensional view of grouping data to provide quick access to strategic information for analysis purposes. In the tourism industry, especially hospitality, this is very useful, especially in processing hotel operational data. Although OLAP technology has been widely applied in the hotel industry in reporting business sales, marketing, and reporting management analysis, there is still little research that discusses customer activity analysis. This study develops an OLAP-based analytical CRM system to analyze customer data and classify it into two main segments: geographically and demographically. Transactional data for hotels that have existed for the last three years are converted into a data warehouse, including the ETL process from the original database to a star schema database. It has a fact table and dimensions. Furthermore, OLAP cube operations are performed and generate customer reports. Testing in this study was carried out by calculating the total data's access speed and accuracy from 100 to 5000 customers, and the estimated value was 45.50 to 80 milliseconds in 30 experiments. For further research, it can be developed by integrating data from the hotel transactional system so that the analytical process can run in real-time.

1. Introduction

Online Analytical Processing (OLAP) is a process used to query data in complex forms and analyze large volumes of data, trend analysis, and structured multidimensional modeling [1]. OLAP capabilities can access various data sources, supports faster and more consistent Query Response Time and data aggregation in the Data Warehouse capable of navigating data in the form of Drill Down and Roll Up [2]. The use of OLAP technology has grown in popularity since 1991. The increase in data volume and the use of business analytics value for enterprises is enormous [3]. Transactions that occur every day must follow the business processes carried out, recorded in a system, and produce data with immense size and are increasing every day [4]. It is stored in a data warehouse, an integrated database architecture archives for processing the data. It analyzes historical data such as customers, sales, and other daily operations information [5]. The extraction process is carried out, and the transformation is carried out by applying various formulas, aggregations, and validations to obtain fact and dimension tables. Then, the data cube was generated by the data warehouse with Multidimensional features [6]. With this, you can see the data conceptually with slicing and dicing techniques to facilitate data analysis inside.

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One of the OLAP technologies that have been carried out by many scientists in Analytical Customer Relationship Management (CRM) is using data warehouse storage of activities obtained from operational data designed to determine the best customer [7]. And other studies examine the data warehouse model designed to manage consumer and purchasing data. The design uses a nine-step methodology with a fact constellation schema modeling consisting of two fact tables: a customer fact table and a purchase fact table [8]. Previous research regarding customer licensing reporting using the OLAP method and tested with 500 data records can access at a speed of 90.70md, resulting in purchasing patterns on customer behavior in choosing services [9][10]. One of the main factors to increase business growth in a company is that sometimes there is a lot of unknown information from potential customers to the company, resulting in the company being unable to analyze the desired potential [11]. Over time, this analysis also helps to refine the prospect's previous business decisions by assessing a customer loyalty for purchasing a product or service [12]. Customer loyalty is one of the keys to achieving maximum profit in the company [13]. Many things that affect customer loyalty include customer satisfaction, such as price [14] and company location [15]. The role of analytical CRM is used to determine the profitability and segmentation of which ones are most profitable over time [16]. Customer segmentation aims to group customers into classes based on their behavior and needs in buying a product or service [17]. With segmentation, business people can more easily identify target consumers [18].

In the tourism industry, especially hospitality, this is very useful, especially in processing hotel operational data. In general, OLAP technology has been widely applied in the hotel industry in sales business reporting, reporting management [19]. However, there is still minimal research that discusses the analysis of customer activity. This study creates an Analytical CRM System with OLAP implementation where data processing is taken from customer data and analyzed to make information that is very useful for the manager.

2. Research Method

The data analysis stages at Hotel Simply Valore to get the software engineering process requirements can be done to see the source and current data flow as shown in Figure 1.

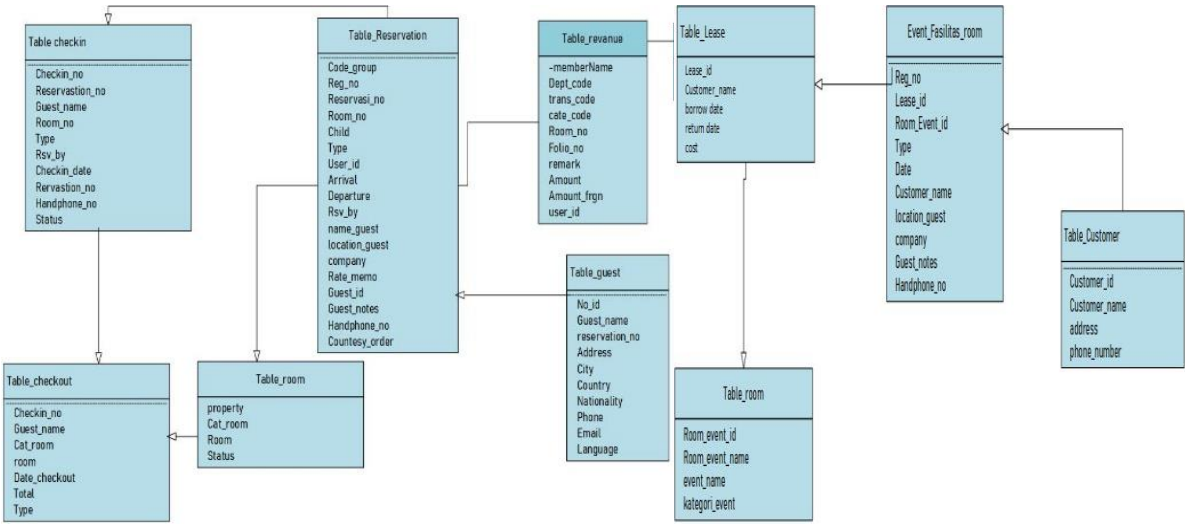


Figure 1. Physical Data Model

After determining the hotel's data source, the next stage will be carried out in Figure 2.

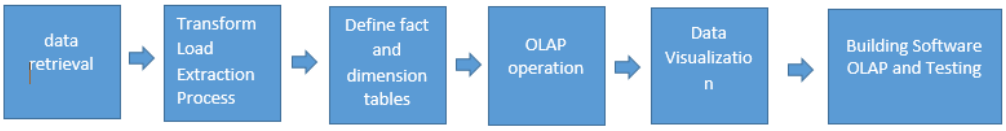


Figure 2. Methodology

2.1 Data Retrieval

Data is taken from hotel transactional activities, which have different sources. The data used are from 2018 to 2020, starting from customer activities such as guest reservation data, event room rentals, and restaurant use in hotels with the xls data format (see Figure 3).

ARRIVAL	GUEST LIS	T (01/	05/18 - 30/12/18)										
No	RmNo	R	esNo	GuestName		RmCat	A/C	Co	NAT	Company/TA	Arrival	Departure	Stay	Email Address	Remark
40	520		36322	Zatadini Nabilah,		DLD3	2/0	0	INA	Booking.Com,	TOUR 26/05/18	27/05/2018	1	0856 9359 0915
41	618		36367	Farhan Ilham,		FMS3	2/0	0	INA	Expedia,	26/05/2018	27/05/2018	1	0895 3979 42509
42	650		36360	Monika,		DLD3	2/0	0	INA	PEGIPEGI.COM,	26/05/2018	27/05/2018	1	
43	1901		36355	LENI,		SPR	2/0	0	INA		26/05/2018	27/05/2018	1	081 387 723 378
44	331		36355	LENI,		DLXS3	2/0	0	INA		26/05/2018	27/05/2018	1	081 387 723 378
45	350		36355	LENI,		DLXS3	2/0	0	INA		26/05/2018	27/05/2018	1	081 387 723 378
46	352		36355	LENI,		DLXS3	2/0	0	INA		26/05/2018	27/05/2018	1	081 387 723 378
47	356		36355	LENI,		DLXS3	2/0	0	INA		26/05/2018	27/05/2018	1	081 387 723 378
48	322		36362	Agus S.		DLXS3	2/0	0	INA	Traveloka.	26/05/2018	27/05/2018	2	0852 202 700 93

Figure 3. Transactional Hotel Data

2.2. Extract Transform Load (ETL)

We perform the ETL process so that data does not experience redundancy, duplicates, inconsistent data, data entry errors [20]. It is selecting and retrieving data from one or several sources (such as databases). Then, we start the data cleaning process by identifying empty (null) data, containing noise, and inconsistent due to the data entry process. Data cleaning waiting time is filled with the maximum appearance value (see Figure 4).

No	NoRoon	NoRes	GuestName	RoomCa	A/C	NAT	Arrival	Departure	Stay	Remark	address
1	626	36203	Achmad Nuzulrullah	FMS3	2/0	INA	16/05/2018	25/05/2018	1	085697258758	Kecamatan Batununggal, Kota Bandung, Jawa Barat
2	522	36208	Yongjun Cai,	DLD3	2/0	INA	20/05/2018	25/05/2018	1	087851423695	Kemayoran,Jakarta Pusat,DKI Jakarta
3	916	36149	Yuriko kayamoto,	DLD3	2/0	INA	23/05/2018	25/05/2018	2	085736262372	Kalideres,Jakarta Barat,DKI Jakarta
4	506	36318	Benediktus dicki kristia	DLD3	2/0	INA	24/05/2018	25/05/2018	4	082330602071	Kecamatan Buahbatu, Kota Bandung, Jawa Barat
5	338	36315	Budi Budiman,	DLXS3	3/0	INA	24/05/2018	25/05/2018	1	083871576645	Kecamatan Cibiru, Kota Bandung, Jawa Barat

Figure 4. Extract data

Data transformation is the process of retrieving raw data that cannot be stored in a data warehouse. Therefore, the data must conform to the data warehouse's predetermined standard structure so that it can be stored in a data warehouse (see Figure 5).

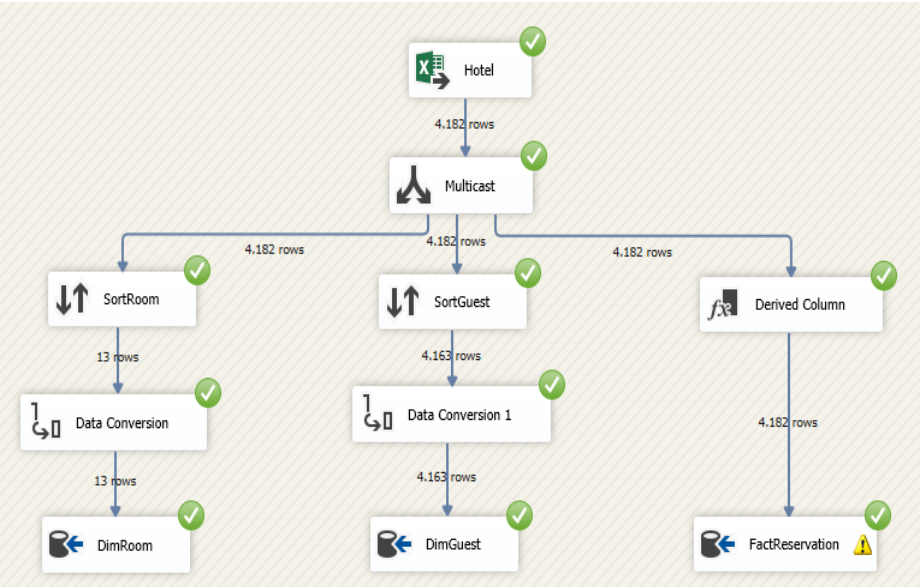


Figure 5. ETL Processes

Load data has been adjusted into the data format made previously, sending data from the Recordset to the destination table in the prepared Data Warehouse. This data can be in the form of dimension data or factual data. Then proceed with the process of filling in the Integration Log data. To be forwarded to the original face and services, where users can access them. Users access the output from this Loading stage in the form of data and information for reporting.

2.3. Define Fact and Dimension Table

As a physical data model, its structure can be seen from various points of view, with three parts (rows, columns, and layers), with multidimensional data objects in it whose attributes are divided into dimensional attributes and measurement attributes (facts) [21]. For dimensional objects in the form of time, which includes the year and month, store customer data in a month or year (see Figure 6). For fact table and dimension, the table can be seen in Table 1 and Table 2.

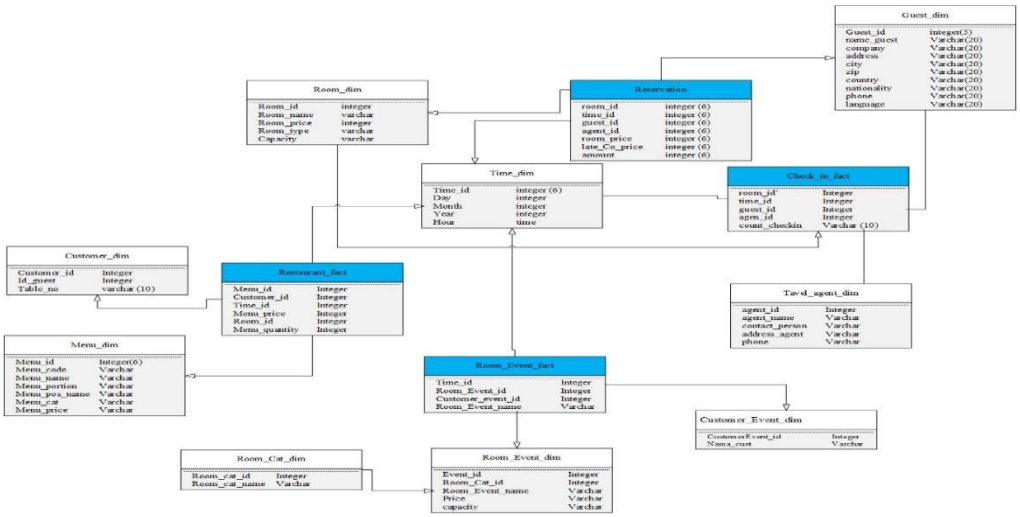


Figure 6. Star Schema

Table 1. Fact Table

Table Fact	Description
Reservation Fact	Serves to see customer characteristics, location, and budget for hotel room selection.
Event Fact	Serves to see event room users in hotels by customers (meeting rooms, weddings, and big conferences).
Agent Travel Fact	Serves to see customers' characteristics, in making reservations to hotels in various ways such as to hotels directly or through applications.
Restaurant Hotel Fact	Serves to see the characteristics of a restaurant in making reservations in various ways, such as to hotels directly or through applications

Table 2. Dimension Table

Table Dimension Name	Attribute	Description
Dimension Guest	GuestId, GuestName, Address, PhoneNumber, City, Region, Gender	This dimension contains information relating to hotel guests
Dimension Room	Roomid, RoomCat, Price, TotalRoom	This dimension describes the room specifications provided by the hotel
Dimension Time	TimeId, Date, Day, month, year	This dimension contains the division of time by year and month. This dimension is useful when management wants to view reports based on a specific time
Dimension CustEvent	CustId, Custname, TypePromosi, CostPromosi	This dimension contains information relating to customers who rent unique meeting rooms
Dimension EventRoom	RoomId, RoomName, Capacity, Price	This dimension describes the specifications of the meeting room facilities provided by the hotel
Dimension Agent	Agent_Id, Agent_name, Agent_email, Agent_number	This dimension contains travel agent information that describes customer transactions when making a reservation
Dimension Restaurant	Restaurant_id, Restaurant_name	This dimension describes the restaurant specifications provided by the hotel
Dimension Menu	Menu_id, Category, Item_name, Price	dimension describes information about the name of the food in the restaurant

2.4. Operation OLAP

Slicing and *Dicing* operations to view data as a visualization of the cube. By slicing and dicing, users can view data from several perspectives. Users can extract part of the aggregated data and can check in detail based on the desired dimensions. Aggregated data is pre-calculated data (pre-calculated). Furthermore, drill-down data aggregation users can do by drill-down and consolidation. Display of more detailed data such as customer information will display customer details.

2.5. Visualization Data

OLAP reporting results are visualized in the form of diagrams or graphs so that the information obtained is easy to understand by displaying information on sales of guest reservations, room rentals, and use of restaurants at Hotel Simply Valore.

2.6. Building Software

Software built for OLAP Analytical CRM reporting using the tools Microsoft SQL Studio and the PHP programming language.

3. Results and Discussion

In this study, an analytical CRM system was built with OLAP implementation to analyze customer data, where there are customer segmentation reporting results in Geography, Demography, and customer travel agents to carry out specific promotions for some hotel customers.

3.1. OLAP In CRM System of Hotel Simply Valore

There is a reservation data analysis menu on the front office dashboard to browse data and determine the number of sales and promotion locations with the most data for management. The decline is based on the data that customers do a lot of activity and recruit new potential customers during hotel promotions**.** The query response time testing can be seen in Table 3.

In Figure 7, the analytical CRM system displays the results of several facts and customer dimensions, including customer reporting each year with this information. The hotel is more aware of each customer's reports every month.

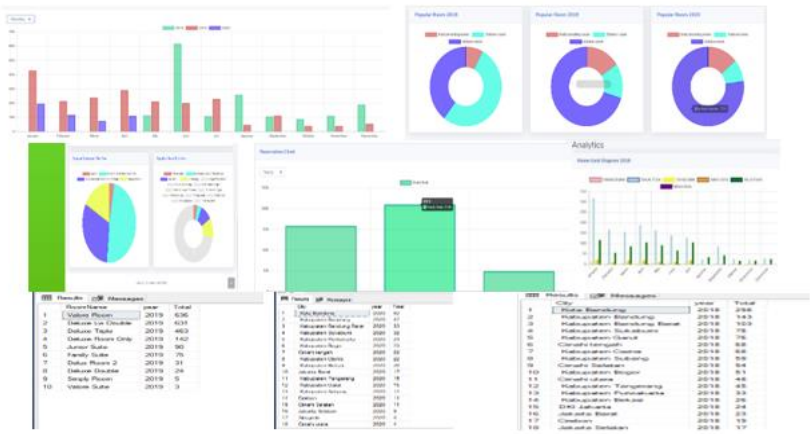
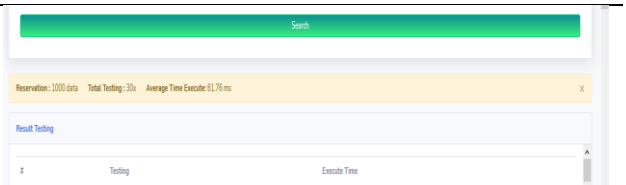


Figure 7. Analytical CRM System

Table 3 Query respond time testing

No	Description	Total Data	Method	Experiments	Average speed (milliseconds)
					OLAP
1	Testing Report on the Number of Hotel Guests (2018-2020)	100	Get	30 Times	44.90
		3000	Get	30 Times	65.77
		5000	Get	30 Times	79.91
2	Testing Report on the Number of Travel Agents (2018-2020)	5000	Get	30 Times	75.00
3	Testing Report Customer Event Room Hotel Facilities (2018-2020)	700	Get	30 Times	49.12
4	Hotel Restaurant Testing Report (2018-2020)	1000	Get	30 Times	53.42
	Result				

The results can be obtained from the query response for data access speeds of less than 80ms to display information from the data in 30 times the experience. With this, the software is feasible to implement. This output is shown by fast data execution and precise accuracy, and a drill-down and roll-up process. Able to analyze more deeply and offer more detailed customer reporting.

4. Conclusions

This study presents an Analytical CRM system in implementing OLAP by displaying information and result data in the form of web-based operable graphs. Olap serves as a useful and essential tool used for analysis needs because of this star schema design. It can analyze data from various dimensional perspectives to quickly capture customer needs. With the use of OLAP technology, data from conventional databases are converted into a data warehouse. Data storage on a large scale so that it can be easily accessed by managers to make decisions. The research found that data access speeds were less than 80ms in 30 times of experience and showed that this software was feasible to implement. This condition was indicated by fast data execution and precise accuracy, and then drill down and roll-up. Able to analyze more deeply. This study produces OLAP Reporting, which can analyze large data sizes and minimize previous software gaps. Faster query response times. Further research can be developed by integrating data from the hotel transactional system so that the analysis process can run in real-time.

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